Packet variants for prime-field ECDLP

Exact checks and the prime-field ECDLP research boundary
30 October 2026 UTC · research/prime_tensor_variants_20261030

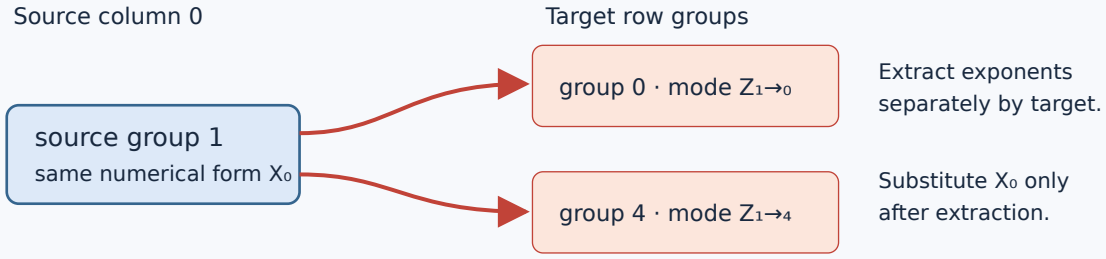
The requested essentially new sub-exponential ECDLP algorithm remains unachieved. This work adapts an existing tensor method to three-adic arithmetic. It supplies an independently reviewed written argument and finite native checks. It does not improve the existing exponent, certify a general implementation or establish novelty.

What changed

The proposed order-three packet action gives real selected-coordinate locality. Independent review found a missing target index: one source tile can cross to two groups. Splitting each mode by target repairs the group factorization and changes the mode bound from `9M/D` to `18M/D`. The retained incorrect formula is an actual negative control, rather than an explanation inferred from timing.

Why the ternary construction needs a target index

Fixed F5 packet: row groups [0,4,1], column groups [1,0,4]



Fixed modular witness: direct = 9, corrected split = 9, faulty unsplit = 0 (mod 27)

packet-unit/MODE-ADDENDUM.md and mode-records/witness.tsv · exact development identity

No timing, global contraction certificate, asymptotic gain or ECDLP novelty is implied.

Mathematical scope

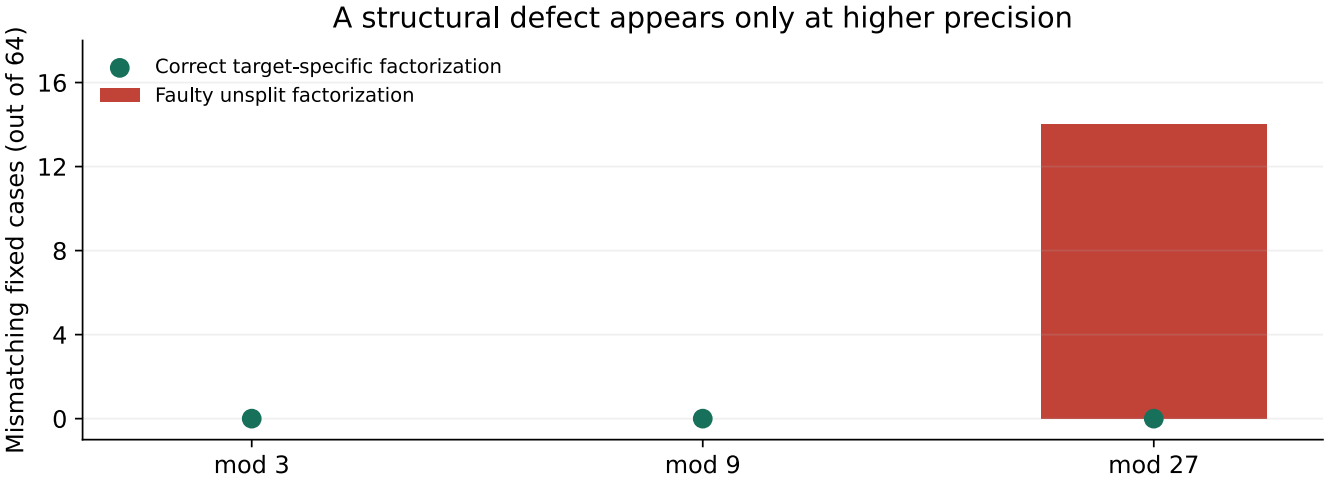
The written adaptation contracts a bipartite Boolean-edge network over powers of 3 in `poly(s+Dlen)` `2^O(s/sqrt(log s))` bit work and polynomial space, under the stated size, degree, precision and charged entry-evaluation bounds. For 0/1 tensors, root precision `M+1` determines the integer count exactly. This is the same exponent class as Appendix D.1 of the pinned October 25 manuscript. Its application to ECDLP still uses the separate published compact hashed-factor-base reduction. This assumes a nonsingular curve over a prime field of characteristic greater than 3 and known odd prime whole order; the source treats a fixed-cofactor extension separately.

Sources and full proof: `sources.md`, `ternary-derivation.md`, `reviews/ternary-global.md`; theorem-to-Lean map: `proof-map.md`.

Exact development evidence

The protocols were frozen before native implementation. These are local correctness checks, with no timing acceptance criteria or scientific isolation claim. Tiny forced parameters do not admit the full asymptotic recursion.

Packet / local identity unit	Recorded count
Full pair classes / orbits	897 / 251
Nonzero slot-locality checks	22,278, all pass
Explicit matching edges covered exactly once	913,731
F67 single-pair locality checks	26,934, all pass
Correct target-split local cases	192 / 192



Population: all 64 three-bit pin pairs at each modulus. Native source: `packet-unit/mode-records/summary.json`; every case row is retained. The graph plots recorded counts, not runtimes or asymptotic extrapolations.

One active call + direct leaves	Recorded count
Root cases at precisions 1 and 2	128 / 128 pass
Mode/direct/exact-chart evaluations	19,200
Actual two-edge child contractions	1,555,200
Explicit child assignments	6,220,800
Full chart coefficient comparisons	19,200
Exact and canonical-lift Hermite recoveries	128 / 128 pass

The unit uses `U^3-U-1` over the three-adic coefficient ring, 25 distinct residue-field lifts, six active Hermite nodes, and child precisions 121 and
122. It explicitly sums all four assignments per two-edge child. A separate

six-bit direct sum and exact rational chart control the factorization.

Flow: all planned fixed cases completed; no case excluded. The first strict one-level build failed on an unused helper; its diagnostic is retained. Two final rebuilds replay byte-identically. Parent replays also pass. Sources: `packet-unit/README.md`, `onelevel-unit/README.md`, both protocols, all case rows, build logs and independent source reviews.

Formalization and remaining gaps

Twenty base Lean theorems check the arbitrary-additive-group rotation/action, selected-coordinate locality, uniform positive reached fibers and the fixed-point hypothesis including F67. A versioned addition supplies eight more theorems: actual integer valuations give the required nonzero digit and a valid bounded slot coordinate. The 28 unique theorems pass explicit module builds, transitive axiom audit, unmodified shared gate and official fresh kernel replay. Only `propext`, `Classical.choice` and `Quot.sound` occur. Base failed attempts and later probe diagnostics are preserved.

Pins: Lean v4.35.0; shared crypto-lean `148f5dc389b341401e06d76abe7783c4ccf42f2e`; Mathlib `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`. Exact hypotheses, commands and remaining gaps are in `lean/PROOF_MAP.md` and `lean-digits/PROOF_MAP.md`. Kernel support, graph packing, three-adic precision, target-split factorization, the recursive machine and complexity are not yet Lean formalized. One active native call does not test multiple active levels.

Other investigated shortcuts

- Stronger mode weights and taper changes do not improve the current enumeration bound by themselves. This is not an arbitrary-algorithm lower bound.
- A direct product deformation restores passive channels. A broader proposed code-occupancy barrier was false without extra premises; repeated-slot counterexamples are retained, and no general impossibility is claimed.
- A concrete scalar window verifier has `O(b^(20/19))` Boolean edges, but the cited contraction bound then gives no sub-exponential guarantee in `b`.
- A sharper heavy-vertex collision formula does not remove whole-vertex padding restrictions. Explicit Boolean code is not automatically an exact weighted multilinear-extension evaluator.
- The literal Fourier graph has large dense binary contraction width. That argument does not exclude implicit methods or a changed representation.

Evidence: `route-decisions.md` and all linked original/independent reviews. These additional written arguments remain outside the completed Lean scope.

Request status and next research obligation

Essential novelty, a complete original algorithm, general implementation audit, exhaustive historical six-repository coverage and isolated increasing-size measurements remain open. No leaderboard or benchmark result is promoted. No performance graph changes: existing scoreboard/timeline and prior-study visuals are left unchanged because no new admitted runtime or pipeline value was measured. This continuation adds its own evidence diagram and count graph.

The selected next bet is a common correlated-packet kernel, described on the next page. Exact compact scalar-predicate splitting remains an alternative. Full requirement mapping: `requirements.md`. Reproduction: `README.md`, `check.sh`, and the pinned Lean project.

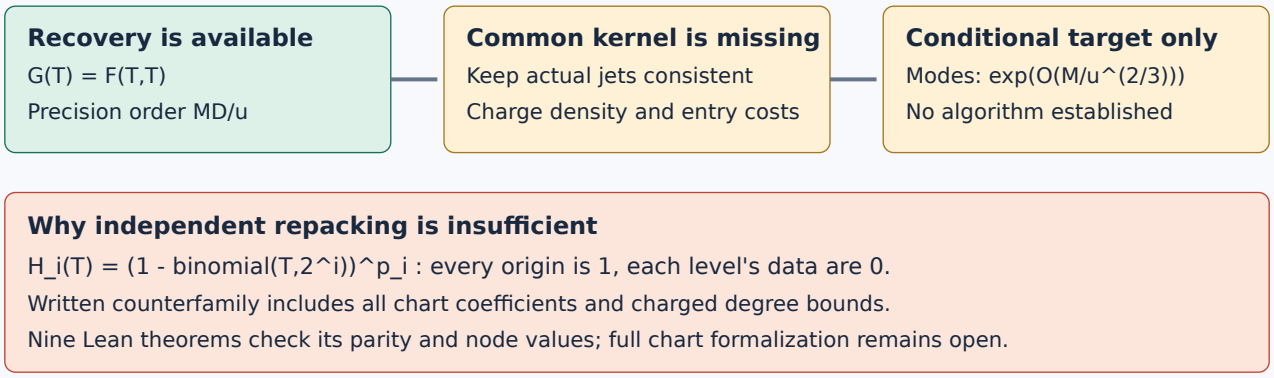
Primary references

Akshima et al., *A Subexponential Time Algorithm for Elliptic Curve Discrete Logarithm*, v0.2, 29 October 2026; source commit `1ee44e562b1b9a50ea339f3b8f81112b5a24b96f`, bochum-crypto/subexponential-ecdl.
The Exponential Time Hypothesis is False, 25 October 2026, Appendix D; source commit `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`, openai/math.
Exact URLs, PDF hashes, scope and literature-search limits: `sources.md`.

The one next bet selected from this round

Try a single deformation with compatible two-axis local factorizations at each valuation level. The reason is concrete: origin recovery has the needed precision already; it is the common geometry and charged mode density that remain missing. This is a structural matrix problem independent of a secret scalar. A change of coefficient prime alone does not supply it. This lead uses the binary source kernel, separately from the ternary units.

Selected next bet: one kernel, compatible local factorizations



Why independent repacking is insufficient

H_i(T) = (1 - binomial(T,2^i))^p_i : every origin is 1, each level's data are 0.

Written counterfamily includes all chart coefficients and charged degree bounds.

Nine Lean theorems check its parity and node values; full chart formalization remains open.

For a product deformation of bidegree `(d,d)`, restricting to `G(T)=F(T,T)` gives degree at most `2d`. The source's Hermite argument recovers the origin from diagonal matched-level samples at precision order `MD/u`. Thus mixed-level samples are unnecessary for this step. Two independent derivations are retained.

The hoped-for two-index mode count `O(M/D^2)` per `(b1,b2)` would give the conditional envelope `exp(O(M/u^(2/3)))`. However, imposing equality at all retained axis bits leaves only `L^2/2^u` physical slots. Direct counting loses the hoped-for density; a constructive finite-support moment reduction restores the old mode-enumeration envelope, without proving a faster algorithm.

> Changing the packet geometry separately by level needs a proved common deformation or congruent-jet identity. Shared origin, bounded degree and integral charts alone do not suffice. No exponent improvement or new ECDLP algorithm is established here.

The written binomial counterfamily satisfies all three generic properties yet has zero levelwise data and origin one. It charges negative nodes, every chart coefficient, degree and denominator sizes. Nine additional checked Lean theorems prove signed binomial parity, rational-polynomial evaluation, origin and node divisibility. They do not prove full coefficientwise chart integrality or the entire data-collision theorem. There are 37 named theorem declarations across the study, rather than a formalized global algorithm.

Actual realization of the counterfamily by repacked Cauchy networks is unknown. It refutes a generic data-only bridge, not every compatible kernel or a code-aware algorithm. A valid new bridge must exploit and prove that additional structure. If every required density, recursion and cost property closed, the inherited prime-field reduction would have a conditional target exponent `O(b/(log b)^(1/3))`. This is a research target, not a runtime bound.

Sources: `reviews/matched-level.md`, `reviews/matched-recovery-independent.md`, `reviews/congruence-moments-independent.md`, `reviews/levelwise-binomial-counterexample.md`, and `lean-binomial/PROOF_MAP.md`. Concrete acceptance conditions: `problems/correlated-packet-common-kernel.md` at the repository root. No timing or native implementation of this proposed common kernel has been run.